

7-Day Sprint Plan

Monday 31 August to Sunday 6 September. Two people, 8–9 hours a day, starting from Agora credentials and nothing else. **Nothing cut** — three personas, multilingual, observer view, all eleven PS11 capabilities.

Day 1 Mon 31 Aug — today

Round 3 ends Thu 3 Sep

Submit Sun 6 Sep, 11 PM IST

Capacity ~120 person-hours

Scope ~110 hours

1 Full scope in seven days — what that costs

Everything ships. The schedule below fits roughly 110 hours of work into roughly 120 available, which means it works — but the slack that absorbs a bad day is gone. Two structural consequences follow, and both are manageable if you plan for them today rather than discovering them on Friday.

1.1 The two consequences of keeping everything

CONSEQUENCE 1 — INTEGRATION SHRINKS FROM A FULL DAY TO HALF A DAY

Feature work now runs through Friday afternoon, so the dedicated integration-and-freeze block becomes **Saturday morning** rather than all of Friday. That is enough *if* you have been running the mock harness continuously and integrating each evening. It is not enough if you leave everything to merge at the end.

The mitigation is non-optional: merge and run a real session every single evening from Day 2, even when it feels like a distraction. Half a day of integration only works when there is nothing large left to discover.

CONSEQUENCE 2 — THREE PERSONAS WILL EXHAUST YOUR AGENT-MINUTES UNLESS YOU ARE DISCIPLINED

Agora bills per **agent**-minute. A ten-minute interview with three personas costs 30 minutes, not 10. Running three personas throughout development would spend roughly 305 of your 300 free minutes before you record a single video take.

The rule for this week: develop and test at `PANEL_SIZE=2` with four-minute sessions. Switch to three personas only for Saturday's integration runs and the video. That single discipline brings the total to ~207 minutes and keeps ~90 in reserve. Make `PANEL_SIZE` an environment variable on Day 2, not later.

1.2 Everything that ships

FEATURE	STATUS	BUILT	NOTE
Interruptible real-time voice	Ships	Day 1	PS11 capability 1 and the mandatory Agora requirement.
Three personas — Priya, Arjun, Meera	Ships	Day 1–2	All three written on Day 1, all three joining on Day 2. Test at two, demo at three.
Floor control & handoff policy	Ships	Day 2–3	Capabilities 3 and 5. Includes the PS11 technical-to-product scenario.
Six conductor tools	Ships	Day 2	Full set, not four — <code>launch_scenario</code> and <code>flag_contradiction</code> are needed for Day 4.
Rubric scoring with evidence quotes	Ships	Day 2–3	Capabilities 4 and 9 from one component.
Vagueness detection	Ships	Day 3	Half of capability 8.
Contradiction detection	Ships	Day 4	The other half, and your most impressive moment.
Difficulty ladder	Ships	Day 4	Capability 7. A judge will test this by answering badly on purpose.
Full scenario library — 3 per persona	Ships	Day 4	Capability 6, at full depth rather than one token scenario.
Cited final report + PDF	Ships	Day 4	Capability 10. No sentence without a citation.
AI disclosure	Ships	Day 3	Capability 11.
Multilingual — Hindi	Ships	Day 5	Organisers' stated interest. Config plus prompt line plus a UI toggle.
Human-in-the-loop observer view	Ships	Day 5	Organisers' stated interest. The largest optional item — a second interface.

1.3 Do these two things before writing code today

- 1. **Message the organisers about the hospitalisation.** The guidelines state the organising team may modify timelines and approve exceptions. One message, real possible upside, no downside. With zero slack in this plan, an extra day would be worth a great deal.
- 2. **Book the Round 3 mentorship session** for tomorrow or Wednesday. Spend it entirely on Agora specifics — silent-agent behaviour, three-agents-in-one-channel patterns, and whether participant credits exist. Being stuck on Agora for a day is the one failure this schedule cannot absorb.

1.4 The two lanes and the one contract

PERSON	OWNS	RESPONSIBLE FOR
Person A Platform	<code>agora/</code> <code>gateway/</code> <code>mock/</code> <code>web/</code>	Agent lifecycle, tokens, the OpenAI-compatible endpoint, SSE, provider fallback, mock harness, both interfaces, deployment.
Person B Intelligence	<code>conductor/</code> <code>state/</code> <code>analysis/</code>	Floor control, handoff policy, ledgers, rubrics, scoring, contradictions, difficulty, scenarios, the report.

AGREE THIS IN THE FIRST TEN MINUTES TODAY

```
# src/contract.py – both people import from here
def next_utterance(session_id: str, role: str, messages: list) -> Iterator[str] | None:
    """None → persona is silent this turn (gateway emits [DONE], no LLM call)
       iter → persona holds the floor; gateway streams chunks to Agora"""
```

A builds the plumbing around it, B builds everything behind it. The mock harness (Day 2) calls the same function, so B can work with Agora switched off. At full scope this matters more than ever — neither of you has time to wait.

2 The Seven Days

Day 1 Mon 31 Aug		Foundation — and prove interruption works before you sleep	
PERSON A — PLATFORM	~9H	PERSON B — INTELLIGENCE	~9H
2h Repo, environment, deps Python 3.11, venv, pinned requirements. Push the skeleton immediately. <pre>pip install fastapi uvicorn httpx openai \ google-genai pyyaml python-dotenv \ agora-token-builder weasyprint pip freeze > requirements.txt</pre> 3h Gateway skeleton gateway/app.py · gateway/sse.pyRoute /v1/{session}/{role}/chat/completions. Hardcode the reply first. Blank line after every SSE frame; [DONE] is mandatory or the agent hangs until timeout. 1h Deploy to Hugging Face Spaces Docker SDK, uvicorn on 7860. curl it from a phone hotspot — if it answers from a network you do not control, Agora can reach it. 3h Agora: join one agent and interrupt it agora/tokens.py · agora/agent.pyUse the Voice Agent Builder console or Agora's sample client — do not build your own UI today. Wrap leave in finally from the very first test.		1h Contract and data models contract.py · state/models.pyFour dataclasses: Turn, Claim, Evidence, session. Every one carries a turn_id — that is what makes citation free on Day 4. 4h Three personas as data conductor/personas.yamlPriya (technical), Arjun (product), Meera (behavioural). Under 150 words each: one distinct concern, an explicit "you interrupt when..." line, one tone example. Arjun's trigger is no_business_framing — that is the PS11 demo moment. 3h Ledgers and session state state/ledger.py · state/session.py<pre>TurnLedger.add(speaker,text)->turn_id ClaimLedger.add / .by_topic(topic) EvidenceLedger.add(...) SessionState: floor_holder, difficulty, ewma, consecutive_turns, rubrics{}, active_scenario, shared_digest(8)</pre> 1h Real next_utterance() Check floor_holder; return None if this role does not hold it, canned text otherwise. A wires the gateway to it tonight.	
TONIGHT — MERGE AND TEST You interrupted an AI mid-sentence and it stopped. If not, message the mentor and Discord immediately — this is the only failure this schedule cannot absorb.		Agora ~10 · total 10	

Day 2 Tue 1 Sep		Three voices, one speaker — and the harness that frees Person B	
PERSON A	~9H	PERSON B	~9H
2h Mock replay harness <code>mock/replay.py</code> Posts a scripted interview in Agora's exact shape, calling <i>every</i> persona per turn, asserting exactly one speaks. Seed with bait: a claim, a contradiction of it, a vague answer, and a strong technical answer with zero customer mention. 2h Wire Groq into the gateway <code>gateway/providers.py</code> Base-URL swap on the OpenAI client. <code>max_tokens=300</code> — terse interviewers are more realistic and it stretches the quota. 2h Three-agent join <code>agora/session.py</code> <code>PANEL_SIZE</code> from env — set it to 2 for development today. Distinct UID and voice per persona, teardown of <i>all</i> agents in <code>finally</code>. 1h Validate the silent-agent path Second agent returns <code>[DONE]</code> immediately — genuine silence, or an artefact? If it fails, fall back to <code>remote_rtc_uids: []</code> plus <code>update</code> on handoff. 2h Tool-call interception Strip tool deltas from the stream, buffer partial JSON per <code>tool_call.id</code>, route to the conductor. A leaked tool call gets read aloud as gibberish — test this deliberately.		2h Conductor v1 — floor control <code>conductor/conductor.py</code> One <code>floor_holder</code>; holder streams, everyone else returns <code>None</code>. Round-robin today. Add a permanent one-speaker assertion to the harness — it protects you all week. 4h All six tools <code>conductor/tools.py</code> <code>request_handoff</code>, <code>flag_vague_answer</code>, <code>flag_contradiction</code>, <code>adjust_difficulty</code>, <code>launch_scenario</code>, <code>record_evidence</code>. <code>strict: true</code>, <code>additionalProperties: false</code>, parse with <code>json.loads</code>. Build all six now — Day 4 needs the last two and there will be no time to add them. 3h Rubric generation <code>analysis/rubrics.py</code> Gemini with <code>response_schema</code>. Five to seven concepts, weights normalised to 10. Extract braces before parsing, and log loudly on fallback — a silent default rubric produces scores that look fine and mean nothing.	
TONIGHT — MERGE AND TEST Two voices in one channel, exactly one speaks, interruptible · <code>python -m src.mock.replay</code> runs a full interview at zero agent-minutes		Agora ~20 · total 30	

PERSON A

~9H

- 5h
Candidate web client
`web/index.html` Agora Web SDK: join, publish mic, subscribe, live transcript, **speaking-now indicator**, persistent AI disclosure banner. Vanilla HTML and JS — no build step to debug on Saturday night.
- 2h
Provider fallback chain
429 → Cerebras → OpenRouter. Log which provider served each turn. A rate limit during the demo is indistinguishable from a crash.
- 1h
AI disclosure in the greeting
Lead persona's `greeting_message` states plainly that all interviewers are AI. One config line, one full PS11 capability.
- 1h
Latency instrumentation
End-of-speech to first audio byte. Log median and p90 — you want a real number for the deck.

PERSON B

~9H

- 3h
Scorer with evidence quotes
`analysis/scorer.py` Ask which rubric concepts were covered **and for the verbatim quote proving each**. Never keyword-match — substring matching marks everything covered and inflates every score toward 10.

```
{"concepts": [{"concept": "sharding", "covered": true, "quote": "shard by tenant id"}], "depth": "moderate", "missing": ["failure modes"]}
```
- 1h
Vagueness detection
Punting-phrase list plus a specificity check — numbers, named tools, concrete examples. Penalise up to 2 points.
- 2h
Handoff policy
`conductor.decide_floor()`

```
pending_handoff → named role  
contradiction → original asker  
tech ok + no biz → product ← PS11  
3 turns held → rotate  
2 vague in a row → hold, pin down  
scenario active → locked to owner
```
- 2h
Shared context digest
Each persona sees a compact summary of what the candidate told the others. Cap at 8–10 turns plus a rolling summary; match Agora's `max_history`.
- 1h
Async analysis worker
`analysis/worker.py` Fire-and-forget after each turn; results inform the *next* turn. Must never block a response — measure latency with it on and off.

MILESTONE A live interview where a technically sound answer with no customer framing makes Priya accept and Arjun challenge — different voice, referencing what was said to Priya. *Record this session as fallback footage.* **Agora ~25 · total 55**

Day 4 Thu 3 Sep		Adaptation, contradictions, scenarios, the report — Round 3 ends today	
PERSON A	~9H	PERSON B	~10H
3h Difficulty meter and tool-call feed in the UI Surface <code>request_handoff</code>, <code>flag_contradiction</code> and flags as they fire. Judges cannot score reasoning they cannot see, and everything the conductor does is invisible by nature. 2h Report view in the browser Render the assessment with citations that jump to the transcript line. A five-second demo moment that proves the evidence chain is real. 2h Scenario UI affordance Show when a role-play is active and who owns the floor, so a viewer understands why one persona is holding it. 2h Start the README Architecture, stack, setup, and a top-level heading "How Agora Conversational AI is integrated" — the organisers list that as its own submission item.		2h Difficulty ladder <code>conductor/difficulty.py</code><pre>ewma = 0.4*score + 0.6*ewma band: <4 easy 4-7 med >7 hard shift after 2 consecutive out-of-band turns (hysteresis, or it oscillates) → persona prompt, UI meter, ledger</pre> Test <i>both</i> directions — a ladder that drops but never recovers is trivially exposed. 3h Contradiction detection <code>analysis/contradiction.py</code> Compare each new claim only against prior claims on the same topic. Handing a small model twenty unrelated claims produces confident nonsense. Require both quotes; bias toward "no conflict" when uncertain — a false accusation on stage is worse than a miss. 2.5h Full scenario library <code>conductor/scenarios.py</code> Three per persona. Entered via <code>launch_scenario</code>, floor locked to the owner until complete — a role-play interrupted after one turn is a non sequitur, not a scenario. 2.5h Cited final report <code>analysis/report.py</code> Per-competency scores, strengths, gaps, hire signal. Rule: no sentence without a citation. WeasyPrint to PDF	
TONIGHT Two weak answers visibly drop the difficulty · the contradiction bait is caught with both quotes · a role-play runs several turns · the report contains no uncited claim		Agora ~25 · total 80	

Day 5 Fri 4 Sep		Multilingual and the observer view — last feature day	
PERSON A	~9H	PERSON B	~9H
5h Human-in-the-loop observer view <code>web/observer.html</code> Live transcript, current floor holder, difficulty level, raised flags — plus three buttons that inject conductor directives: <i>push harder</i>, <i>move on</i>, <i>hand to Product</i>. The conductor already accepts directives, so this is mostly interface. 2h Multilingual — Hindi Set the ASR language or enable multilingual recognition; pick TTS voices with the matching locale; add a UI toggle so it can be demonstrated deliberately rather than hoped for. 2h Deploy and cold-test Push everything, then open the link on a device that has never run the project, on a network that is not yours.		1h Persona language mirroring One line in each prompt: reply in the language the candidate used. Test that a mid-answer switch does not break the shared context digest. 2h Observer directive handling Wire the three buttons into the conductor as injected directives that influence the next turn without breaking floor control. 3h First full integration pass Run two complete interviews end to end at <code>PANEL_SIZE=2</code>. Fix what breaks. This is what makes Saturday's half-day enough. 3h Persona tuning The highest-value hours of the week. Tighten prompts until all three sound genuinely distinct and stay terse. On free-tier models this affects the experience more than any architectural change left available.	
FEATURE COMPLETE A human can nudge the panel mid-interview · a candidate can answer in Hindi and the panel continues · two clean end-to-end runs		Agora ~30 · total 110	

Day 6 Sat 5 Sep

Freeze at noon, then assets — the tightest day in the plan

MORNING — BOTH, INTEGRATION & FREEZE ~4H • Three full interviews at <code>PANEL_SIZE=3</code> Different topics and different candidate personalities — a confident candidate and a hesitant one exercise completely different paths through the difficulty ladder and the vagueness detector. • Walk the PS11 list against a live session All eleven capabilities observed happening, not "the code exists". Only the first one scores. • Break it deliberately Kill the network mid-interview, exhaust a rate limit, confirm the fallback chain engages and teardown still runs. Check for orphaned agents after every test. • Freeze at noon. Anything unfinished is cut, not rushed. From here the only work is assets.	AFTERNOON — SPLIT ~5H • Person B — record the demo video Two or three takes with API keys untouched all week. Show, do not narrate. 0:00 disclosure spoken + shown 0:20 Priya opens, distinct voice 0:45 follow-up quotes the answer 1:15 CANDIDATE INTERRUPTS 1:35 handoff fires on screen 1:50 Arjun challenges the business case 2:20 contradiction caught, both quotes 2:45 difficulty meter drops 3:05 Meera runs a role-play 3:25 candidate switches to Hindi 3:40 recruiter clicks "push harder" 3:55 cited report renders 4:15 latency + architecture • Person A — README, architecture doc, deck Finish the Agora integration section, the build disclosure (built 31 Aug – 6 Sep, point at commit history), licence check, and fold the measured latency figures into the deck.
TONIGHT All eleven capabilities verified live · video recorded · README and deck drafted. If the video is not done, it is Sunday morning's first job. Agora ~90 · total 200	

Day 7 Sun 6 Sep		Buffer, then submit — deadline is 11 PM, aim for 4 PM	
MORNING — FINISH AND VERIFY ~5H		AFTERNOON — SUBMIT ~2H	
• Whatever slipped from Saturday This morning exists because the plan has no slack. If everything is done, you gain five hours; if not, you are not in trouble. • Deck final pass and rehearsal Round 4 includes a live online presentation, not just an upload. Run it twice, timed. • Verify every link while logged out Repo public, deployment reachable, video playable. <i>"Incomplete or inaccessible submissions may not be considered."</i>		• Submit by 4 PM IST Seven hours before the deadline. Deadline-day submissions lose to upload failures and portal timeouts every year. • The checklist Name and description · PS11 · both members' details · repo · deployment link · demo video · architecture doc · Agora integration section · pitch deck · build disclosure. • Then stop. No features can be added after Round 4 anyway. 8–11 Sep is polish and rehearsal only.	
DONE Submitted with seven hours to spare · ~100 agent-minutes still in reserve		Agora ~10 · total 210	

3 Running a plan with no slack

At full scope there is no spare day. That does not make the plan reckless — it makes the checkpoints load-bearing. Each night you either pass the checkpoint or you make a scope decision. What you must not do is carry a failed checkpoint forward silently.

3.1 Nightly discipline — the three rules

1. **Merge and run a real session every evening.** Not the harness — an actual Agora session, even four minutes. Saturday's half-day integration only works if nothing large is left to discover.
2. **Develop at `PANEL_SIZE=2`, demo at 3.** This single habit is the difference between finishing with 90 agent-minutes spare and running out on Saturday morning.
3. **Miss a checkpoint, make a decision that night.** Either fix it the next morning before starting new work, or pull the trigger on §3.3. Never carry it forward hoping to catch up — with no slack, that is how a plan fails on the last day rather than the first.

3.2 Where this plan can fail

RISK	IMPACT	WHAT TO DO
Interruption does not work tonight	Fatal	Message Discord and the mentor tonight. Person B's entire lane continues via the mock harness, so only half the team is blocked while it is resolved.
Empty completions do not silence an agent	High	Tested Day 2. Fallback: <code>remote_rtc_uids: []</code> plus subscription swaps via <code>update</code> on handoff.
Day 3 web client overruns	High	The largest single block in the plan and the most likely overrun. If it slips past 6h, ship a transcript-only page — the speaking indicator and disclosure banner matter, the styling does not. Recover the rest on Day 4.
Day 5 runs out of hours	Medium	The observer view is the last thing built for exactly this reason. Cut it to a read-only dashboard with no buttons — you keep the human-in-the-loop story, lose only the interaction.
Agent-minutes run low before Saturday	High	Check the running total every night against the figures in this plan. If you are ahead of budget by Day 4, drop test sessions to three minutes and move all debugging to the harness.
Rate limit during the video	High	Fallback chain built Day 3. Reserve untouched keys for Saturday's recording. Give each person separate dev keys.
Health setback	Medium	The lanes are independent. If one person loses a day, the other continues and you trigger §3.3 rather than trying to catch up.

3.3 The contingency list — not a plan, a lever

Nothing here is scheduled to be cut. This is the order to cut in *if* a checkpoint fails and you need hours back. Pull from the top.

#	CUT THIS	RECOVERS	WHAT YOU LOSE
1	Observer view interactivity	~3h	Nothing scored. Keep the read-only dashboard and the story survives intact.
2	Multilingual	~3h	A bonus capability, not a PS11 line.
3	Scenarios: 3 per persona → 1	~1.5h	Nothing — capability 6 still ships, at less depth.
4	Third persona (develop and demo at 2)	~2h + 60 agent-min	Panel feels smaller, but PS11 says "one or more" and the technical-to-product moment still lands.
5	Report PDF export	~1.5h	Keep the browser view; drop WeasyPrint.
6	Contradiction detection	~3h	Half of capability 8. Vagueness detection still covers the other half. Cut this only as a last resort — it is your most impressive moment.

Below item 6 everything is load-bearing. If you are cutting further than that, shorten the demo instead — a four-minute interview that runs flawlessly beats a ten-minute one that stutters.

THE THREE THINGS THAT MUST BE TRUE BY WEDNESDAY NIGHT

1. A candidate can interrupt an agent mid-sentence and it yields.
2. Personas with distinct voices take turns, and exactly one speaks.
3. A technically sound answer with no customer framing triggers a handoff from Priya to Arjun.

Those three are the submission. Everything after Wednesday makes it better; nothing after Wednesday can replace them. If Wednesday ends without all three, stop adding features on Thursday and fix them instead — then re-enter the plan at Day 4 with the contingency list open.

YOU ARE IN BETTER SHAPE THAN IT FEELS

You qualified for Round 3 while one of you was in hospital, which means the idea and the framing already convinced a screening panel. The Agora credentials — the one thing that cannot be rushed — are done. Full scope is roughly 110 hours against 120 available, and the plan is sequenced so neither of you is ever waiting on the other.

It is tight, and it will only stay on track if the nightly checkpoints are honoured. But it fits.